

Information Sheet for a Regular Paper

Same Words, Different Procedure: Counterfactual Transfer Tests for Reusable LLM-Agent Skills

1. What is the main claim of the paper, and why is it important to the autonomous-agents and multi-agent-systems literature?

Success after supplying an agent with a reusable skill is not, by itself, evidence that the skill caused success. A base agent may already solve the target, generic grounding and search controls may explain the gain, or a lexically similar source may encode an incompatible procedure. The paper’s main claim is that a state-matched, six-condition counterfactual protocol can separate these explanations and attribute target success specifically to procedural compatibility under the tested executor.

The Counterfactual Skill Transfer Test (CSTT) crosses compatible versus incompatible source procedures with near versus far lexical similarity on the same native target. It adds a no-skill condition and a binding-only condition that receives the same target grounding and search safeguards but no source procedure. This distinction matters for agent architecture evaluation: without it, a skill library can be credited for behavior supplied by the base model or the executor. CSTT also produces deployment-relevant measures of helpful, harmful, redundant, and shared-failure transfer. The contribution fits the journal’s interests in evaluating agent decision-making architectures, testbeds for agent analysis, and software-engineering methods for autonomous agents.

2. What evidence supports the claim?

After two retained failed pilots and one final mechanism revision, the complete method, source assignment, three model revisions, inference procedure, decision gates, and a zero-overlap confirmation roster were frozen. Confirmation uses 60 previously untouched ALFWorld targets, three open local model families, six conditions per target–model block, 180 paired blocks, and 1,080 native episodes.

The procedurally matched but lexically far condition succeeds in 176/180 blocks (97.8%), compared with 18/180 (10.0%) for the lexically near procedural lure. The paired gain is 87.8 percentage points, with a target-stratified 95% bootstrap interval of 82.8–93.3 points and a task-clustered randomization $p = 1.00 \times 10^{-6}$. Relative to the fair binding-only control, matched-far produces 150 helpful and zero harmful discordances, an 83.3-point net gain. The primary effect is 88.3, 88.3, and 86.7 points in the three model families and positive in all six task families. Independent audits accept all 1,080 episodes and 46,662 native turns with exact within-block initial-state matching, valid native actions, valid source assignments, and no target-plan leakage.

The claim is deliberately bounded. The confirmation covers instance-level transfer in one embodied text environment and three small-to-medium open models. It does not establish universal retrieval quality, cross-domain transfer, or benefits for proprietary frontier models.

3. Which papers make the most closely related contributions, and how does this paper differ?

Taylor and Stone, “Transfer Learning for Reinforcement Learning Domains: A Survey,” *Journal of Machine Learning Research* 10 (2009), 1633–1685, and da Silva and Costa, “A Survey on Transfer Learning for Multiagent Reinforcement Learning Systems,” *Journal of Artificial Intelligence Research* 64 (2019), 645–703, establish transfer criteria, baselines, and the importance of source–target relationships. Tuynman and Ortner, “Quantification of transfer in reinforcement learning via regret bounds for learning agents,” *Autonomous Agents and Multi-Agent Systems* 40 (2026), article 16, quantifies transfer relative to learning without shared experience. CSTT applies the same comparative principle to reusable LLM-agent procedures, but uses paired native outcomes and a control that isolates executor assistance.

The closest recent skill studies are Zhou et al., “Counterfactual Trace Auditing of LLM Agent Skills” (arXiv:2605.11946), which labels behavioral influence patterns in paired with/without-skill traces; Yang et al., “SkillMaster” (arXiv:2605.08693), which uses counterfactual probe utility to train skill editing; and Belikova et al., “Managing Procedural Memory in LLM Agents” (arXiv:2606.23127), which benchmarks transfer across tasks, roles, and models. CSTT differs by experimentally crossing procedural compatibility with lexical similarity on the same initial state and by separating source-procedure credit from target binding and generic search controls.

4. Have parts of this paper been published before?

No part of this manuscript has been published in an archival journal or conference, and this manuscript is not under consideration elsewhere. The authors will release the code, records, and manuscript-supporting artifacts in a public repository and a versioned Zenodo archive for reproducibility; these artifact releases are not prior archival publication of the article.

For transparency, a related manuscript by the same authors, “From Traces to Witnessed Contracts: Incremental Precondition–Effect Compilation for LLM Agent Skills,” is currently under consideration at *Automated Software Engineering*. That work studies pre-execution detection of defective skill revisions. The present manuscript studies causal attribution of skill-transfer benefits. The two studies use disjoint confirmation targets and different methods, estimands, experiments, figures, tables, endpoints, and results. They share only the broader agent-skill lifecycle motivation, a public benchmark platform, and general experimental infrastructure. A copy of the related manuscript is supplied separately for editorial assessment.